

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir Praktikum Jaringan Komputer

FIREWALL DAN NAT

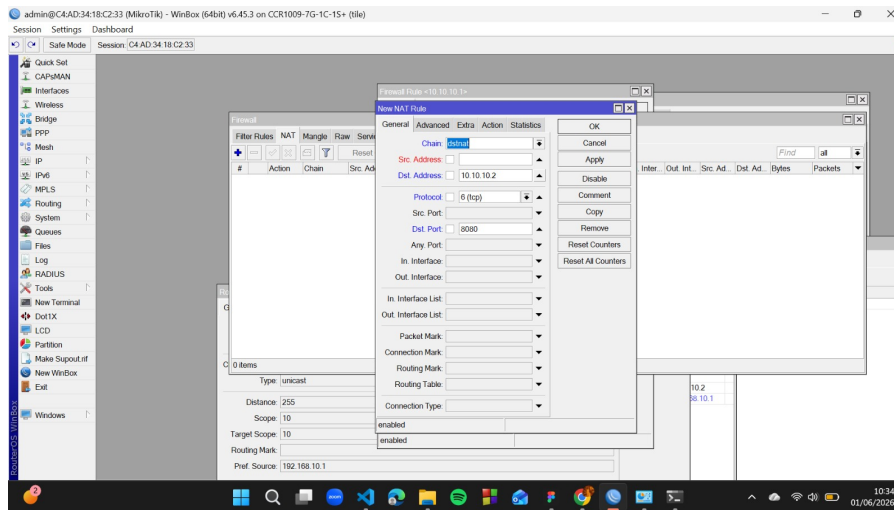
Mathew William Junior Tumanggor - 5024231076

2026

1 Langkah-Langkah Percobaan

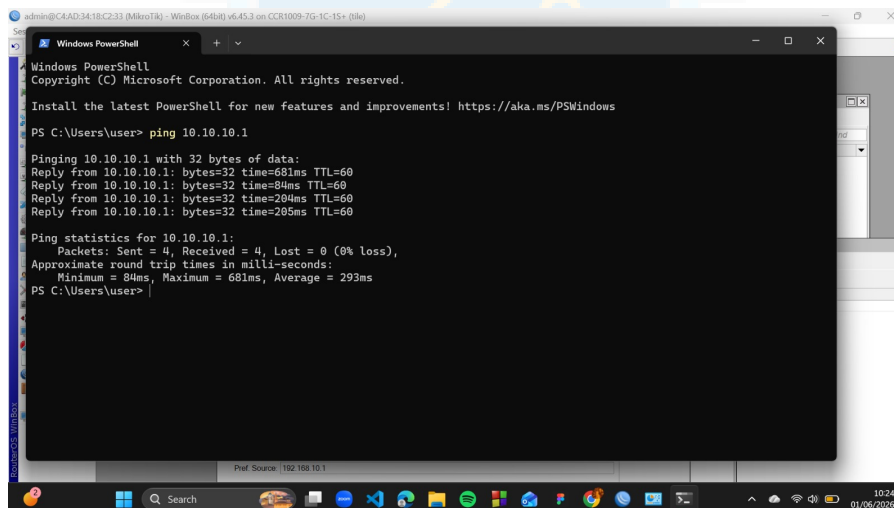
1.1 Praktikum 1: Firewall NAT Dasar

1. **Reset Konfigurasi Router:** Buka aplikasi Winbox untuk mengakses router. Masuk ke menu System → Reset Configuration, centang *No Default Configuration*, lalu klik tombol *Reset Configuration* untuk memastikan perangkat dalam kondisi bersih.



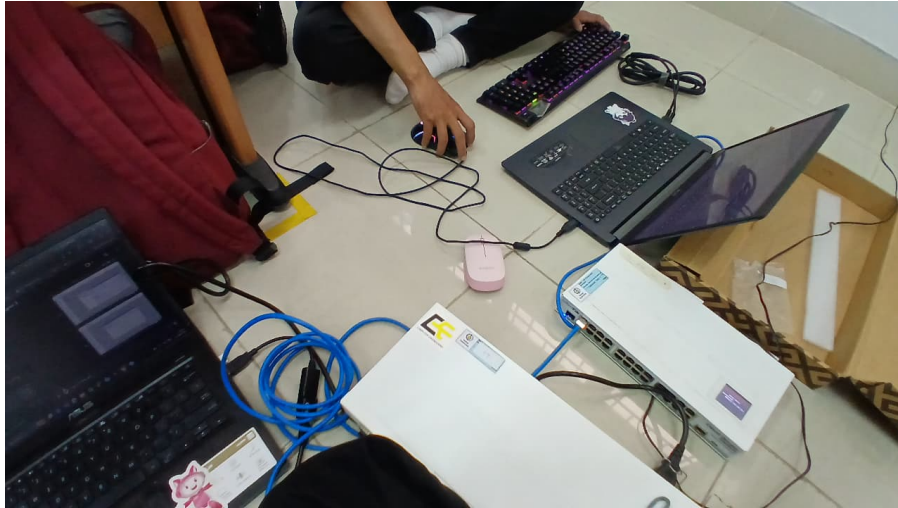
Gambar 1: Proses Reset Konfigurasi Router

2. **Konfigurasi DHCP Client di Router A:** Sambungkan port *ether1* Router A ke jaringan Lab/Internet. Masuk ke menu IP → DHCP Client, klik tombol + (Add), pilih *ether1* pada kolom *Interface*, klik *Apply* lalu *OK*, dan pastikan status berubah menjadi *bound*.



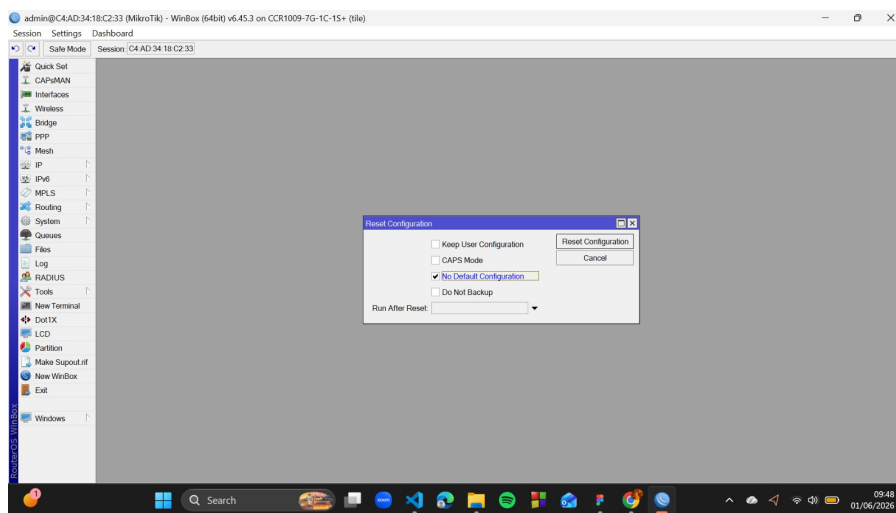
Gambar 2: Konfigurasi DHCP Client pada Interface ether1 Router A

3. **Konfigurasi IP Address di Router A:** Tambahkan IP Address pada *ether2* sebagai jalur penghubung ke Router B. Masuk ke menu IP → Addresses, klik + (Add), masukkan alamat *10.10.10.1/30*, dan pilih interface *ether2*.



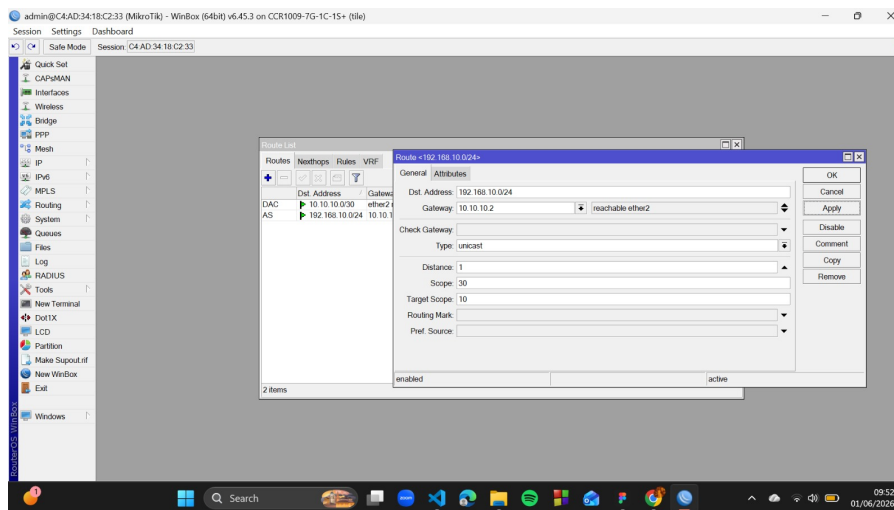
Gambar 3: Penambahan IP Address pada ether2 Router A

4. **Konfigurasi Routing Statis di Router A:** Agar Router A dapat mengenali jaringan lokal PC Client di bawah Router B, masuk ke menu IP → Routes, klik + (Add), isi *Dst. Address* dengan 192.168.10.0/24, dan *Gateway* dengan 10.10.10.2.



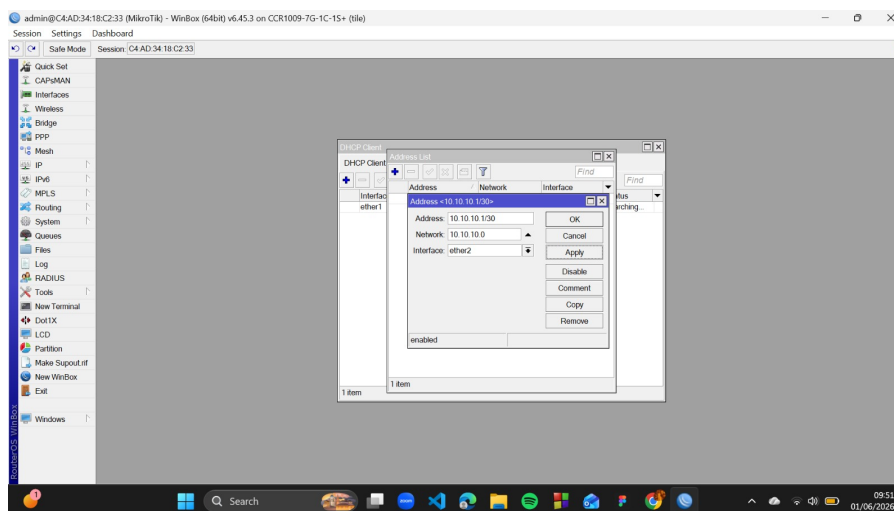
Gambar 4: Konfigurasi Static Route Menuju Jaringan Lokal Router B

5. **Konfigurasi IP Address di Router B:** Buka Winbox Router B. Masuk ke menu IP → Addresses, tambahkan alamat 192.168.10.1/24 pada interface ether2 (ke arah PC Client) dan alamat 10.10.10.2/30 pada interface ether1 (ke arah Router A).



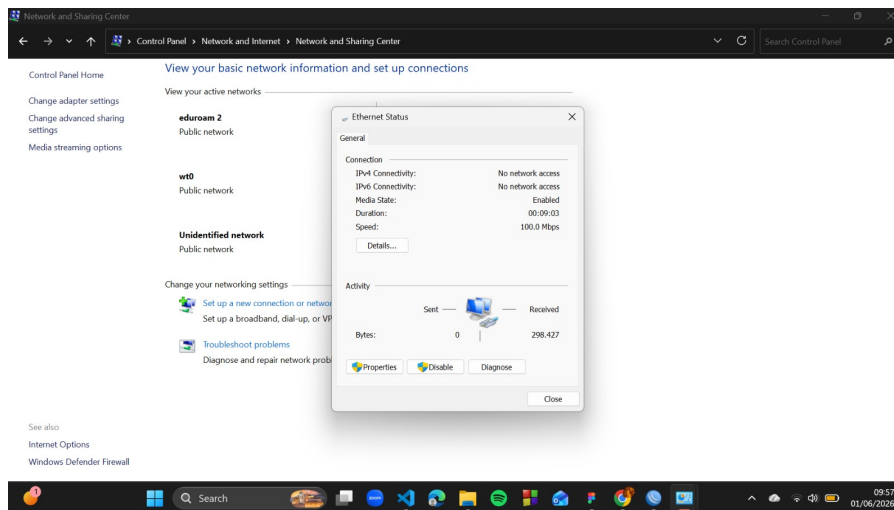
Gambar 5: Konfigurasi Alamat IP ether1 dan ether2 di Router B

6. **Konfigurasi Default Route di Router B:** Masuk ke menu IP → Routes pada Router B, klik + (Add), biarkan *Dst. Address* 0.0.0.0/0, lalu arahkan *Gateway* ke IP 10.10.10.1 agar semua lalu lintas data ke internet diteruskan ke Router A.



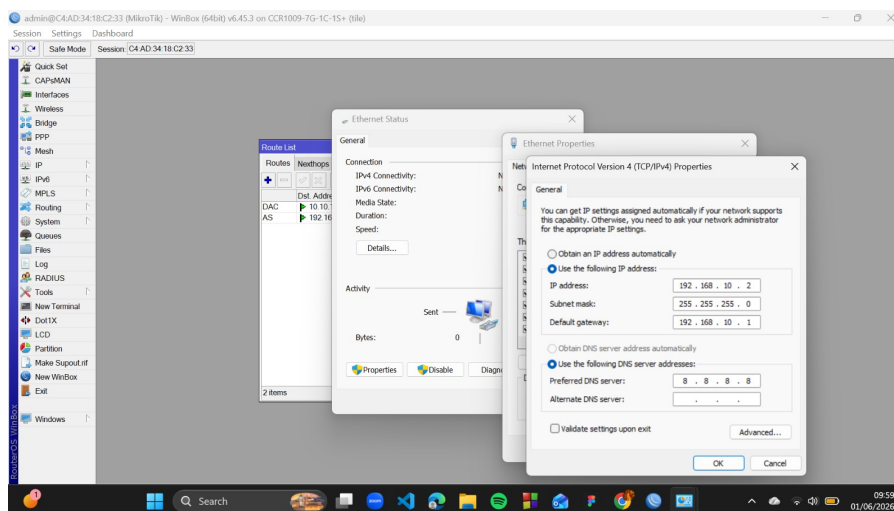
Gambar 6: Pengaturan Default Route pada Router B

7. **Konfigurasi Firewall NAT di Router B:** Masuk ke menu IP → Firewall → tab NAT di Router B. Tambahkan aturan baru dengan *Chain*: srcnat, *Out. Interface*: ether1, dan pada tab *Action*, ubah menjadi masquerade.



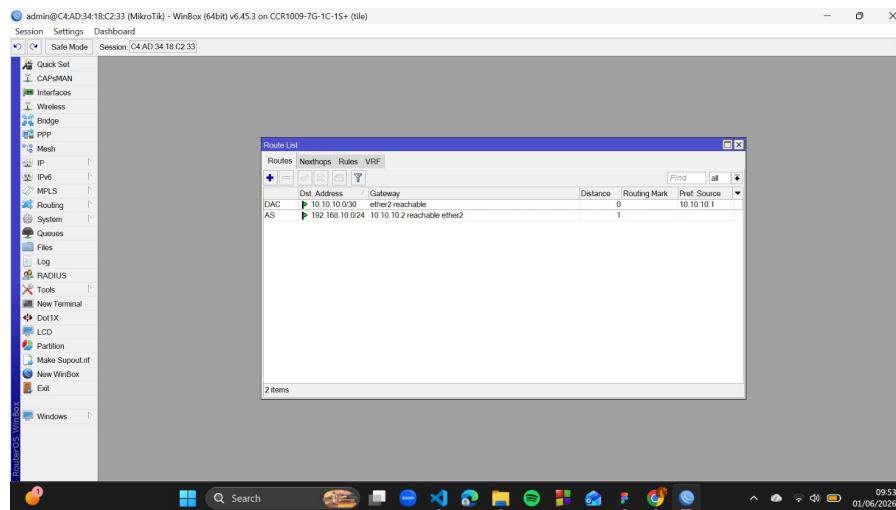
Gambar 7: Konfigurasi Masquerade pada Router B

8. **Konfigurasi IP Statis PC Client:** Atur IP statis pada *adapter network* PC Client dengan parameter IP: 192.168.10.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.1, serta DNS Server: 8.8.8.8.



Gambar 8: Pengaturan Alamat IP Statis pada PC Client

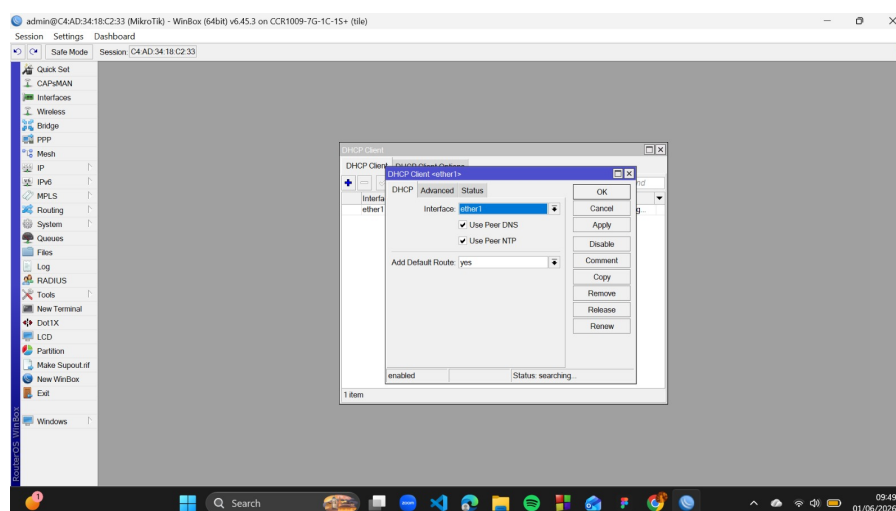
9. **Pengujian Jaringan dari PC Client:** Buka *Command Prompt* (CMD) di PC Client, lakukan pengujian konektivitas menggunakan perintah `ping` berturut-turut ke gateway lokal, gateway Router A, hingga ke alamat IP Internet untuk memastikan enkapsulasi NAT berhasil.



Gambar 9: Hasil Pengujian Ping dari PC Client

1.2 Praktikum 2: Firewall Filter (Input vs Forward)

1. **Konfigurasi Firewall Filter - Chain Input:** Buka Winbox Router B, masuk ke IP → Firewall → tab *Filter Rules*. Tambahkan aturan baru dengan *Chain*: input, *Src. Address*: 10.10.10.1 (Router A), *Protocol*: icmp, dan set *Action* menjadi drop.
2. **Pengujian Jaringan setelah Chain Input:** Akses *New Terminal* pada Router A. Lakukan ping ke IP Router B (10.10.10.2) yang akan menghasilkan *Request Time Out* (RTO), lalu lakukan ping ke PC Client (192.168.10.2) yang akan tetap menghasilkan *Reply*.
3. **Konfigurasi Firewall Filter - Chain Forward:** Nonaktifkan (*disable*) aturan *input* sebelumnya. Buat aturan baru dengan *Chain*: forward, *Src. Address*: 10.10.10.1, *Protocol*: icmp, dan set *Action* menjadi drop.
4. **Pengujian Jaringan setelah Chain Forward:** Lakukan uji ping kembali dari terminal Router A. Hasil koneksi menuju Router B kini menjadi *Reply*, sedangkan uji ping yang melintasi Router B menuju PC Client berubah menjadi *Request Time Out* (RTO).



Gambar 10: Hasil Pengujian Paket ICMP pada Chain Input dan Forward

1.3 Praktikum 3: Firewall NAT Forwarding (Port Forwarding)

1. **Konfigurasi Destination NAT (DstNAT):** Pastikan seluruh filter rules sebelumnya dimatikan. Di Winbox Router B, buka IP → Firewall → tab *NAT*. Tambahkan aturan dengan *Chain*: *dstnat*, *Dst. Address*: 10.10.10.2, *Protocol*: *tcp*, *Dst. Port*: 8080. Pada tab *Action*, set *Action*: *dst-nat*, *To Addresses*: 192.168.10.2, dan *To Ports*: 80.
2. **Inisiasi Web Server Lokal:** Buka *Command Prompt* pada PC Client. Jalankan perintah eksekusi modul *HTTP server* bawaan python melalui instruksi: `python -m http.server 80` untuk mengaktifkan layanan *web server* sementara.
3. **Pengujian Port Forwarding dari Router A:** Buka *New Terminal* di Router A, lalu gunakan *tool fetch* untuk mengunduh halaman web via IP publik Router B dengan perintah `/tool fetch url="http://10.10.10.2:8080" keep-result=no`. Amati proses *download* yang berhasil tanpa adanya indikasi error.
4. **Monitoring Connection Tracking:** Buka menu IP → Firewall → tab *Connections* pada Router B untuk melihat bagaimana sistem mencatat translasi alamat dari IP 10.10.10.1 menuju 10.10.10.2:8080 dengan status koneksi *established*.

2 Hasil Tugas Modul

Berikut adalah alur implementasi teknis, manajemen kebijakan keamanan (*security policy*), serta proses verifikasi jaringan *multi-vendor* pada lingkungan PNET Lab:

1. **Perancangan Arsitektur Jaringan Tiga Zona:** Menginisialisasi dan menghubungkan seluruh *node* infrastruktur enterprise yang memisahkan area publik (*External WAN*), area privat lokal (*Internal LAN*), dan zona server terisolasi (*Demilitarized Zone / DMZ*).

```

Linux5 x Mikrotik x vIOS x Linux6 x

( '>' )
/) TC (\   Core is distributed with ABSOLUTELY NO WARRANTY.
(/-__-__-\)   www.tingcorelinux.net

gns3@box:~$ ping google.com
ping: bad address 'google.com'
gns3@box:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
ping: sendto: Network is unreachable
gns3@box:~$
gns3@box:~$ startx
-sh: startx: not found
gns3@box:~$ ping google.com
ping: bad address 'google.com'
gns3@box:~$ sudo ifconfig eth0 172.16.100.10 netmask 255.255.255.0
gns3@box:~$ sudo route add default gw 172.16.100.1
gns3@box:~$ echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
nameserver 8.8.8.8
gns3@box:~$ _

```

Gambar 11: Skema Topologi Jaringan Enterprise Multi-Vendor

2. **Alokasi Pengalamatan IP Sisi Edge (MikroTik ISP):** Melakukan pemetaan dan aktivasi alamat IP pada interface *backbone* MikroTik ISP yang mengarah ke sisi FortiGate (*ether2*) serta segmen Client WAN (*ether3*) guna memastikan interkoneksi awal Layer 3 berjalan lancar.

```

Linux5 x vIOS x Linux6 x Linux x Mikrotik x Fortinet x

Router con0 is now available

Press RETURN to get started.

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*****

Router>
Router>en
Router#ping 10.20.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#ping 10.20.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#ping 192.168.20.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 8.8.8.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 22/22/23 ms
Router#

```

Gambar 12: Daftar Penetapan Alamat IP pada Interface MikroTik

3. **Konfigurasi Parameter Interface Gateway FortiGate (Fase 1):** Memasuki subsistem *system interface* pada perangkat FortiGate via CLI untuk memberikan alamat IP 10.10.10.2/30 pada port1 (WAN) sekaligus membuka izin akses manajemen jaringan berupa paket ping, HTTPS, SSH, dan HTTP.

```
Fortinet >
set type physical
set snmp-index 3
next
edit "port4"
set vdom "root"
set type physical
set snmp-index 4
next
edit "naf.root"
set vdom "root"
set type tunnel
set src-check disable
set snmp-index 5
next
edit "l2t.root"
set vdom "root"
set type tunnel
set snmp-index 6
next
edit "ssl.root"
set vdom "root"
set type tunnel
set alias "SSL VPN interface"
set snmp-index 7
next
edit "fortilink"
set vdom "root"
set fortilink enable
set ip 10.255.1.1 255.255.255.0
set allowaccess ping fabric
set type aggregate
set lldp-reception enable
set lldp-transmission enable
set snmp-index 8
next
end
```

Gambar 13: Konfigurasi Atribut Keamanan dan IP port1 FortiGate

4. **Konfigurasi Parameter Interface Gateway FortiGate (Fase 2):** Melanjutkan pendefinisian alamat IP internal pada port2 (Jalur INSIDE ke Cisco: 10.20.20.1/30) dan port3 (Jalur DMZ ke Ubuntu: 192.168.20.1/24) untuk mengaktifkan seluruh gerbang keamanan antar-zona.

```
Fortinet >
edit "ssl.root"
set vdom "root"
set type tunnel
set alias "SSL VPN interface"
set snmp-index 7
next
edit "fortilink"
set vdom "root"
set fortilink enable
set ip 10.255.1.1 255.255.255.0
set allowaccess ping fabric
set type aggregate
set lldp-reception enable
set lldp-transmission enable
set snmp-index 8
next
end

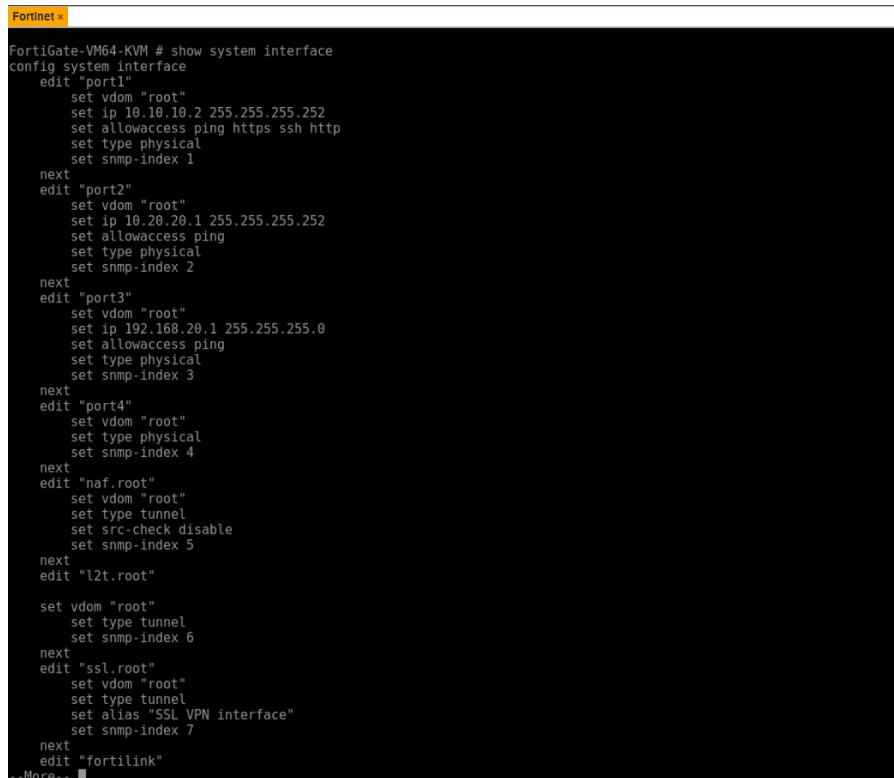
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # get router info routing-table
Command fail. Return code -9999

FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
V - BGP VPNv4
* - candidate default

Routing table for VRF=0
S* 0.0.0.0/0 [10/0] via 10.10.10.1, port1, [1/0]
C 10.10.10.0/30 is directly connected, port1
C 10.20.20.0/30 is directly connected, port2
S 192.168.10.0/24 [10/0] via 10.20.20.2, port2, [1/0]
C 192.168.20.0/24 is directly connected, port3
```

Gambar 14: Konfigurasi Alokasi IP port2 dan port3 FortiGate

5. **Inspeksi Tabel Routing pada Core Firewall:** Mengeksekusi instruksi `get router info routing-table all` melalui terminal FortiGate untuk memastikan rute statis (*static route*) menuju segmen LAN (192.168.10.0/24) telah terdaftar dengan *next-hop* Cisco Router.



```
Fortinet >
FortiGate-VM64-KVM # show system interface
config system interface
  edit "port1"
    set vdom "root"
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping https ssh http
    set type physical
    set snmp-index 1
  next
  edit "port2"
    set vdom "root"
    set ip 10.20.20.1 255.255.255.252
    set allowaccess ping
    set type physical
    set snmp-index 2
  next
  edit "port3"
    set vdom "root"
    set ip 192.168.20.1 255.255.255.0
    set allowaccess ping
    set type physical
    set snmp-index 3
  next
  edit "port4"
    set vdom "root"
    set type physical
    set snmp-index 4
  next
  edit "naf.root"
    set vdom "root"
    set type tunnel
    set src-check disable
    set snmp-index 5
  next
  edit "l2t.root"
    set vdom "root"
    set type tunnel
    set snmp-index 6
  next
  edit "ssl.root"
    set vdom "root"
    set type tunnel
    set alias "SSL VPN interface"
    set snmp-index 7
  next
  edit "fortilink"
```

Gambar 15: Visualisasi Tabel Routing Aktif pada Sistem FortiGate

6. **Penyusunan Kebijakan Akses Firewall (Aturan Internal):** Merancang *security policy* awal pada FortiGate, meliputi izin translasi alamat dinamis (LAN_to_WAN dengan NAT *enabled*) serta izin komunikasi internal menuju server (LAN_to_DMZ dengan NAT *disabled*).

```

Fortinet #
set uuid 5ecc58ec-61db-51f1-8ee6-6ef74dd1f901
set srcintf "port2"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "WAN_to_DMZ_HTTP"
set uuid 7e4d7b24-61db-51f1-8794-89372b97dd6f
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "DMZ_HTTP"
set schedule "always"
set service "HTTP"
next
edit 4
set uuid c9bc69f8-61db-51f1-6715-a1957f3c7890
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "DMZ_HTTP"
set schedule "always"
set service "HTTP"
next
edit 10
set name "DMZ to WAN"
set uuid 60cae3b8-61de-51f1-82b2-d162f651d696
set srcintf "port3"
set dstintf "port1"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
end
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #

```

Gambar 16: Daftar Kebijakan Keamanan Inter-Zone Lokal FortiGate

7. **Penyusunan Kebijakan Akses Firewall (Aturan Publik):** Menambahkan regulasi *traffic filter* eksternal (WAN_to_DMZ_HTTP) yang mengizinkan paket dari luar masuk ke area DMZ secara terbatas, hanya untuk protokol HTTP dengan memanfaatkan objek alamat tujuan.

```

Fortinet #
FortiGate-VM64-KVM # show firewall policy
config firewall policy
edit 1
set name "LAN to WAN"
set uuid 18a161a8-61db-51f1-1830-80edd1d665bc
set srcintf "port2"
set dstintf "port1"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN to DMZ"
set uuid 5ecc58ec-61db-51f1-8ee6-6ef74dd1f901
set srcintf "port2"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "WAN_to_DMZ_HTTP"
set uuid 7e4d7b24-61db-51f1-8794-89372b97dd6f
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "DMZ_HTTP"
set schedule "always"
set service "HTTP"
next
edit 4
set uuid c9bc69f8-61db-51f1-6715-a1957f3c7890
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "DMZ_HTTP"
set schedule "always"
set service "HTTP"
--More--

```

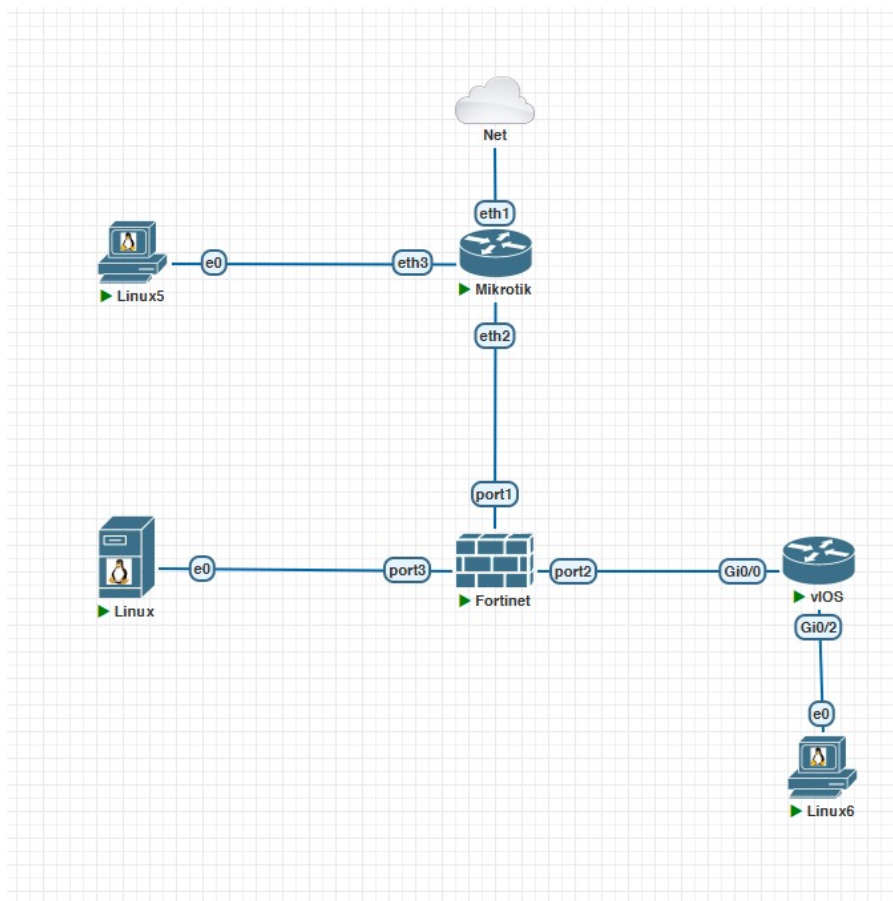
Gambar 17: Spesifikasi Security Policy Akses Web dari Zona Luar

8. **Implementasi Destination NAT (Virtual IP):** Mengonfigurasi modul *Port Forwarding* melalui Virtual IP (VIP) bernama DMZ_HTTP. Mekanisme ini memetakan IP publik FortiGate (10.10.10.2) port 80 menuju IP privat server (192.168.20.10) port 80 secara transparan.

```
FortiGate-VM64-KVM # show firewall vip
config firewall vip
edit "DMZ_HTTP"
set uuid aa29eb24-61db-51f1-a831-e49142b976fb
set extip 10.10.10.2
set mappedip "192.168.20.10"
set extintf "port1"
set portforward enable
set extport 80
set mappedport 80
next
end
```

Gambar 18: Konfigurasi Pemetaan Virtual IP (VIP) Port Forwarding

9. **Uji Validasi Konektivitas Lintas Sektor dari Core Firewall:** Menjalankan uji coba pengiriman paket data menggunakan perintah `execute ping` dari FortiGate menuju gateway ISP, Cisco Router, Ubuntu Server, hingga menguji fungsionalitas DNS internet publik (8.8.8.8).



Gambar 19: Hasil Uji Coba PING Multi-Directional dari Sisi FortiGate

10. **Konfigurasi Logika Interface pada Cisco vIOS Router:** Mengonfigurasi parameter *network layer* pada Cisco Router dengan menetapkan IP 10.20.20.2 pada interface Gi0/0 (arah WAN) dan IP 192.168.10.1 pada Gi0/2 (arah LAN Lokal) disertai instruksi `no shutdown`.

```

Fortinet >
config firewall vip
edit "DMZ_HTTP"
set uuid aa29eb24-61db-51f1-a831-e49142b976fb
set extip 10.10.10.2
set mappedip "192.168.20.10"
set extintf "port1"
set portforward enable
set extport 80
set mappedport 80
next
end

FortiGate-VM64-KVM # execute ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1): 56 data bytes
64 bytes from 10.10.10.1: icmp_seq=0 ttl=64 time=0.5 ms
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.3 ms
^C
--- 10.10.10.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.3/0.4/0.5 ms

FortiGate-VM64-KVM # execute ping 10.20.20.2
PING 10.20.20.2 (10.20.20.2): 56 data bytes
64 bytes from 10.20.20.2: icmp_seq=0 ttl=255 time=1.8 ms
64 bytes from 10.20.20.2: icmp_seq=1 ttl=255 time=0.7 ms
^C
--- 10.20.20.2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.7/1.2/1.8 ms

FortiGate-VM64-KVM # execute ping 192.168.20.10
PING 192.168.20.10 (192.168.20.10): 56 data bytes
64 bytes from 192.168.20.10: icmp_seq=0 ttl=64 time=0.5 ms
64 bytes from 192.168.20.10: icmp_seq=1 ttl=64 time=0.2 ms
^C
--- 192.168.20.10 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.2/0.3/0.5 ms

FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=111 time=22.1 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=111 time=22.0 ms
^C
--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 22.0/22.0/22.1 ms

```

Gambar 20: Status Interface dan Pengalamatan IP pada Cisco vIOS

11. **Penyelarasan Network Profile Jaringan Internal (Client LAN):** Mengonfigurasi parameter IP statis 192.168.10.10/24 pada interface `eth0` Client LAN Tinycore (Linux6) serta menambahkan rute *default gateway* agar mengarah ke interface Cisco Router.

```

Mikrotik x Fortinet x Linux5 x Linux x Linux6 x vIOS x

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* of the IOSv Software or Documentation to any third party for any *
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*****

Router>
Router>
Router>
Router>
Router>show ip interface brief
Interface          IP-Address      OK? Method Status      Pr
otocol
GigabitEthernet0/0  10.20.20.2      YES manual  up          up
GigabitEthernet0/1  unassigned      YES unset   administratively down do
wn
GigabitEthernet0/2  192.168.10.1    YES manual  up          up
GigabitEthernet0/3  unassigned      YES unset   administratively down do
wn
Router>show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.20.20.2      YES manual  up          up
GigabitEthernet0/1  unassigned      YES unset   administratively down  down
GigabitEthernet0/2  192.168.10.1    YES manual  up          up
GigabitEthernet0/3  unassigned      YES unset   administratively down  down
Router>

```

Gambar 21: Konfigurasi Alamat IP Statis dan Default Route Client LAN

12. **Penyelarasan Network Profile Jaringan Eksternal (Client WAN):** Mengonfigurasi profil *adapter network* `eth0` pada Client WAN Tinycore (Linux5) dengan menyuntikkan IP 172.16.100.10, netmask 255.255.255.0, serta rute *gateway* utama ke arah MikroTik ISP.

```

Linux5 x Mikrotik x vIOS x Linux6 x

( '>')
/) TC (\   Core is distributed with ABSOLUTELY NO WARRANTY.
(/-__-__-\)      www.tinycorelinux.net

gns3@box:~$
gns3@box:~$ http://192.168.20.1
-sh: http://192.168.20.1: not found
gns3@box:~$ sudo ifconfig eth0 192.168.10.10 netmas 255.255.255.0
ifconfig: bad address 'netmas'
gns3@box:~$ sudo ifconfig eth0 192.168.10.10 netmask 255.255.255.0
gns3@box:~$ sudo route add default gw 192.168.10.1
gns3@box:~$ echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
nameserver 8.8.8.8
gns3@box:~$

```

Gambar 22: Konfigurasi Alamat IP Statis dan Default Route Client WAN

13. **Penyusunan Lingkungan Web Server pada Zona DMZ:** Melakukan penyetelan IP statis 192.168.20.10 di Ubuntu Server, melakukan instalasi paket aplikasi Nginx, serta memodifikasi struktur file index.html menggunakan teks identitas resmi kelompok praktikan.

```

Mikrotik x Fortinet x Linux5 x Linux6 x vIOS x

0 10.10.10.2 timeout
1 10.10.10.2 timeout
2 10.10.10.1 84 64 986ms host unreachable...
3 10.10.10.2 timeout
4 10.10.10.2 timeout
5 10.10.10.2 timeout
6 10.10.10.2 timeout
7 10.10.10.2 timeout
8 10.10.10.2 timeout
9 10.10.10.2 timeout
10 10.10.10.2 timeout
sent=11 received=0 packet-loss=100%

[admin@MikroTik] > ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 D 10.4.89.101/24 10.4.89.0 ether1
1 10.10.10.1/30 10.10.10.0 ether2
2 172.16.100.1/24 172.16.100.0 ether3
[admin@MikroTik] > ping 10.10.10.2
SEQ HOST SIZE TTL TIME STATUS
0 10.10.10.2 56 255 0ms
1 10.10.10.2 56 255 0ms
sent=2 received=2 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=0ms

[admin@MikroTik] > ip address print detail
Flags: X - disabled, I - invalid, D - dynamic
0 D address=10.4.89.101/24 network=10.4.89.0 interface=ether1
actual-interface=ether1

1 address=10.10.10.1/30 network=10.10.10.0 interface=ether2
actual-interface=ether2

2 address=172.16.100.1/24 network=172.16.100.0 interface=ether3
actual-interface=ether3
[admin@MikroTik] >

```

Gambar 23: Status Layanan Nginx dan Modifikasi Berkas HTML Ubuntu DMZ

14. **Verifikasi Konektivitas Akhir Jaringan Internal (Cisco Router):** Mengeksekusi pengetesan stabilitas jaringan akhir melalui *command prompt* Cisco Router menuju interface INSIDE Forti-Gate, IP internal Ubuntu DMZ, hingga ke server publik internet luar.

```
Linux5 * Mikrotik * VIOS * Linux8 * Linux *
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
Active: active (running) since Sun 2026-06-07 04:42:38 CST; 6h ago
Docs: man:nginx(8)
Process: 611 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Process: 632 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
Main PID: 633 (nginx)
Tasks: 3 (limit: 4620)
Memory: 14.6M
CGroup: /system.slice/nginx.service
├─633 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
├─634 nginx: worker process
└─635 nginx: worker process

Jun 07 04:42:37 kom systemd[1]: Starting A high performance web server and a reverse proxy server...
Jun 07 04:42:38 kom systemd[1]: Started A high performance web server and a reverse proxy server.
root@kom:~# ip addr show eth0
2: eth0: <BRDADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:5b:1e:00:bb:00 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::525b:1eff:fe00:bb00/64 scope link
        valid_lft forever preferred_lft forever
root@kom:~# ip addr add 192.168.20.10/24 dev eth0
Error: either "local" is duplicate, or "0" is a garbage.
root@kom:~# ip addr add 192.168.20.10/24 dev eth0
root@kom:~# ip route add default via 192.168.20.1
root@kom:~# echo "nameserver 8.8.8.8" > /etc/resolv.conf
root@kom:~# apt update && apt install -y nginx
Hit:1 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal InRelease
Get:2 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal-updates InRelease [128 kB]
Get:3 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal-backports InRelease [128 kB]
Get:4 https://mirrors.tuna.tsinghua.edu.cn/ubuntu focal-security InRelease [128 kB]
Fetched 383 kB in 3s (137 kB/s)
Reading package lists... Done
Building dependency tree
Reading state information... Done
313 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree
Reading state information... Done
nginx is already the newest version (1.18.0-0ubuntu1.7).
0 upgraded, 0 newly installed, 0 to remove and 313 not upgraded.
root@kom:~# echo "Tunod_4_DMZ_Firewall_21-KelompokFiberOptic" > /var/www/html/index.html
-bash: /var/www/html/index.html: No such file or directory
root@kom:~# echo "Tunod_4_DMZ_Firewall_21-KelompokFiberOptic" > /var/www/html/index.html
root@kom:~# systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable nginx
root@kom:~# systemctl start nginx
```

Gambar 24: Hasil Uji Validasi Konektivitas ICMP Melalui Cisco Router

Tugas modul ini mengimplementasikan skenario jaringan skala enterprise dengan arsitektur *multi-vendor* yang memisahkan area publik (WAN), jaringan lokal (LAN), dan zona demilitarisasi (DMZ). Jaringan dirancang secara modular dengan menempatkan firewall FortiGate sebagai pengendali lalu lintas utama (*central stateful firewall*) untuk mengatur kebijakan keamanan dan segmentasi jaringan secara ketat menggunakan *object policy*.

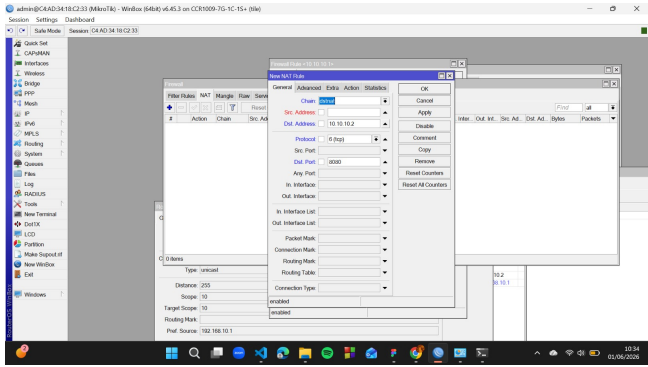
Implementasi *Virtual IP (VIP) / Port Forwarding* pada FortiGate (IP 10.10.10.2) memungkinkan *Client WAN Tinycore* untuk mengakses layanan *Web Server Nginx* pada *Ubuntu Server DMZ* (192.168.20.10) melalui protokol HTTP port 80. Seluruh fungsionalitas pemetaan port dan restriksi paket ICMP dari zona luar (WAN) berhasil diverifikasi melalui kombinasi pengujian utilitas *ping* dan penelusuran protokol HTTP. Berdasarkan hasil pengujian tersebut, kebijakan keamanan berhasil menjamin bahwa layanan web server publik di dalam DMZ dapat diakses oleh pihak luar secara aman tanpa mengekspos topologi maupun *resource* privat di area LAN. *Client WAN* dibatasi secara ketat hanya untuk akses web tanpa diberikan izin untuk melakukan *ping* ke alamat internal DMZ maupun LAN di balik *Cisco Router*, sehingga integritas data dan keamanan jaringan dalam tetap terjaga secara otomatis.

3 Kesimpulan

Praktikum Modul 4 ini berhasil memberikan pemahaman komprehensif mengenai fungsionalitas dan konfigurasi sistem keamanan jaringan berbasis *Firewall Filter* dan NAT. Praktikan mampu mengklasifikasikan penggunaan *Chain Input* untuk mengamankan *resource* internal router serta *Chain Forward* untuk mengontrol distribusi paket antar-subnet. Di samping itu, pengenalan *Source NAT* dan *Destination NAT* memberikan solusi praktis dalam mengatasi keterbatasan alokasi IPv4 global serta teknik publikasi server lokal secara aman melalui bantuan tabel *Connection Tracking*.

4 Lampiran

4.1 Dokumentasi saat praktikum



Gambar 25: Dokumentasi Kegiatan Praktikum Firewall dan NAT